

SMART INDIA HACKATHON 2026 • BACKEND & GIS SPEC

GIS Interactive Dashboard — Developer Manual

A step-by-step implementation guide for building the live Flood Disaster Risk & Relocation Map with Antigravity AI.

ROLE Dashboard Dev (GIS)	TARGET DISTRICT Majuli, Assam	MAP CENTER [26.95°N, 94.20°E]	BACKEND REGION asia-south1 (Mumbai)
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■■ Note on ML & Hazard Data: Hazard scores and sub-factors are currently mock demo data from Data/ML team. The backend schema is finalized and will remain identical when live ML models replace it.

1. Feature Overview & Requirements

Feature Component	What It Does in Simple Terms	Live Function / Source
Base Interactive Map	Leaflet/MapLibre map centered on Majuli district with terrain/dark tiles.	React Leaflet
Village Markers	8 village pins plotted by lat/lng, color-coded by flood hazard priority.	httpsCallable('getVillages')
Inspector Side Drawer	Clicking a village displays hazard sub-factors, population, & scores.	httpsCallable('getVillageDetail')
Safe Relocation Sites	Shield markers showing safe high-plinth evacuation zones with capacities.	Firestore relocation_sites
Safe Route Line	Draws a highlighted dashed line connecting village to best safe site.	httpsCallable('getSiteMatches')
Filter Controls	Tabs/Chips to filter map by Immediate, Short-term, Medium, Monitor.	getVillages({ priority_category })

2. Risk Categories & Color Palette

Category	Score Cutoff	Pin Color	Hex Code	Tailwind CSS
■ Immediate	Score >= 71.0	Vibrant Red	#EF4444	bg-red-500 text-white
■ Short-term	Score 51.0 - 70.9	Orange	#F97316	bg-orange-500 text-white
■ Medium-term	Score 31.0 - 50.9	Amber / Yellow	#EAB308	bg-yellow-500 text-black
■ Monitor	Score < 31.0	Emerald Green	#10B981	bg-emerald-500 text-white
■■ Safe Relocation Site	Candidate Site	Blue / Cyan Shield	#3B82F6	bg-blue-600 text-white

3. Step-by-Step Implementation Guide with Antigravity AI

Step 1: Install Packages & Configure Firebase SDK

In your frontend app (Vite / React), install leaflet, react-leaflet, and firebase:

```
npm install leaflet react-leaflet lucide-react firebase
# Create src/firebase.js with region: 'asia-south1'
```

■ **Prompt to Antigravity:** "Initialize Firebase SDK in src/firebase.js using getFunctions(app, 'asia-south1'). Set up a responsive React dashboard layout with Tailwind CSS and React-Leaflet."

Step 2: Center Base Map on Majuli, Assam

Initialize Leaflet centered at [26.95, 94.20] with zoom 11. Use CartoDB Positron / Voyager raster tiles for high-contrast river basin geography.

```
<MapContainer center={[26.95, 94.20]} zoom={11} className='h-full w-full'>
<TileLayer url='https://{s}.basemaps.cartocdn.com/rastertiles/voyager/{z}/{x}/{y}{r}.png' />
</MapContainer>
```

Step 3: Plot Color-Coded Village Markers

Fetch villages on load via getVillages(). Create custom Leaflet L.divIcon pins colored by priority_category (with a pulsing ring on Immediate pins).

■ **Prompt to Antigravity:** "Create a MapPins component that calls getVillages from Firebase. Render custom circular SVG markers on lat/lng with color coding: Immediate (#EF4444 with pulse animation), Short-term (#F97316), Medium-term (#EAB308), Monitor (#10B981)."

Step 4: Village Inspector Panel (Side Drawer)

When a user clicks a village pin, open a sliding drawer displaying:

- **Village Header:** Name, District, Priority Badge (Score & Category)
- **Demographics:** Population, Elderly %, Road Access tier
- **Explainability Breakdown:** Top 3 hazard factors from top_factors array with percentage bars
- **Action CTA:** 'Find Optimal Relocation Site' button

■ **Prompt to Antigravity:** "Build a sliding right-side drawer component for the selected village. Show priority score gauge, category badge, demographics, and an explainability section with progress bars for top 3 hazard factors."

Step 5: Safe Relocation Sites & Safe Route Line

When a village is clicked, call getSiteMatches({ villageId }) to rank candidate sites. Render blue shield markers for relocation sites and draw a dashed <Polyline> linking the village to the #1 recommended safe site.

■ **Prompt to Antigravity:** "When a village is selected, call getSiteMatches(villageId). Plot candidate sites with blue shield markers. Draw a dashed cyan Polyline from the village to the top-ranked site with a distance tooltip."

Step 6 & 7: Filter Controls, Hazard Toggles, & Loading States

- **Filter Chips:** All (8), Immediate (2), Short-term (2), Medium-term (2), Monitor (2).
- **Hazard Overlays:** Layer switcher for Landslide Risk Heatmap and Rainfall contours.
- **UX Polish:** Semi-transparent loading spinner while data loads; error retry banner on network fault.

4. Backend API Contract Quick Reference

Callable Function	Input Arguments	Key Output Structure
getVillages	{ district?, priority_category? }	{ count: 8, villages: [{ id, name, lat, lng, priority_score, priority_category, top_factors }] }
getVillageDetail	{ villageld: string }	{ village: { id, name, population, hazard_factors, top_factors, recommended_site } }
getSiteMatches	{ villageld: string }	{ village_id, matches: [{ site_id, site_name, distance_km, suitability_score, criteria_breakdown }] }
updateWeights	{ priority: {...}, site_ranking: {...} }	{ message, new_weights, villages_recomputed: 8 }